



YAML

What is YAML?

- YAML is a Data Serialization Language.
- A New Language for Data
- Modeled after Perl-like structures
- But not just intended for Perl
- Human friendly, cross language, Unicode based data serialization language.

What is YAML?

- YAML can be taken as an acronym for either
 - **Yet Another Markup Language**
 - **YAML Ain't Markup Language**
- Like JSON, the purpose of YAML is to represent typical data types in human- readable notation

Why YAML if XML?

- Unlike XML which is not easily readable by humans, YAML was created to be *human-friendly* and integrate easily with modern programming languages.
- Unlike with XML, YAML was intended to simplify the viewing and understanding of *config files, log files, object persistence, and messaging*, to allow the programmer to spend more time programming and less time formatting data.

YAML Design Goals

YAML design goals:

- YAML documents are very readable by humans.
- YAML interacts well with scripting languages.
- YAML uses host languages native data structures.
- YAML has a consistent information model.
- YAML enables stream-based processing.
- YAML is expressive and extensible.
- YAML is easy to implement.

Design consideration

- To achieve Human/Computer easy processing
- High Data/Markup (Signal/Noise) Ratio
- To be the world's best Data::Dumper
- To be great for Configs, Caches, and Logs
- Unicode based
- Speedy parsing
- Alternative to XML (for some applications)

Basic YAML Syntax Rules

- Documents begin with --- and end with ...
- Indentation of lines denotes the structure within the document.
- Comments begin with #
- Members of lists begin with –
- Key value pairs use the following syntax <key>:value>
- YAML files should end in .yaml
- YAML is case sensitive.
- YAML does not allow the use of tabs. Spaces are used instead as tabs are not universally supported.

YAML Syntax Basics

- Collections
- Streams and Documents
- Mappings (hashes / dictionaries)
- Sequences (arrays / lists)
- Comments

Contd..

- Scalars (strings,integers, dates and other atomic data types)
 - Simple
 - Quoted
 - Block (|)
 - Folded
 - Wiki
 - Escaping
- Anchors & Aliases (&,*)
- Tags (Data types)("!")
- Nodes -YAML *nodes* have *content* of one of three *kinds*: scalar, sequence, or mapping. In addition, each node has a *tag* which serves to restrict the set of possible values which the node's content can have.

Collections:

YAML's block collections use indentation for scope and begin each entry on its own line.

Example 2.1. Sequence of Scalars
(ball players)

```
- Mark McGwire  
- Sammy Sosa  
- Ken Griffey
```

Example 2.4. Sequence of Mappings
(players' statistics)

```
-  
  name: Mark McGwire  
  hr: 65  
  avg: 0.278  
-  
  name: Sammy Sosa  
  hr: 63  
  avg: 0.288
```

Structures – Streams & Documents

YAML uses three dashes (“---”) to separate *documents* within a *stream*. Three dots (“...”) indicate the end of a document without starting a new one

Example 2.7. Two Documents in a Stream
(each with a leading comment)

```
# Ranking of 1998 home runs
---
- Mark McGwire
- Sammy Sosa
- Ken Griffey

# Team ranking
---
- Chicago Cubs
- St Louis Cardinals
```

Example 2.8. Play by Play Feed
from a Game

```
...
time: 20:03:20
player: Sammy Sosa
action: strike (miss)
...
time: 20:03:47
player: Sammy Sosa
action: grand slam
...
```

Structures – Streams & Documents

- A question mark and space (“? ”) indicate a complex mapping key.
- & and * indicate Anchors and Aliases

Example 2.10. Node for “Sammy Sosa” appears twice in this document

```
---  
hr:  
  - Mark McGwire  
    # Following node labeled SS  
  - &SS Sammy Sosa  
rbi:  
  - *SS # Subsequent occurrence  
  - Ken Griffey
```

Example 2.11. Mapping between Sequences

```
? - Detroit Tigers  
  - Chicago cubs  
:  
  - 2001-07-23  
  
? [ New York Yankees,  
    Atlanta Braves ]  
: [ 2001-07-02, 2001-08-12,  
    2001-08-14 ]
```

Mappings

- A YAML mapping is like a Perl hash
- Unordered Key/Value pairs
- Separated by ':' (space is mandatory)

name: Benjamin

rank: Private

serial number: 1234567890

12:34 PM: My favorite time

Sequences

- A YAML sequence is like a Perl array
- An ordered collection of data
- YAML has a bullet like syntax '- '

- red
- white
- blue
- pink

Examples

Fruits:

- Apples
- Tomatoes

Veggies:

- Spinach
- Broccoli

Meats:

- Burgers
- Shrimp

Household:

- Candles
- Incense
- Toilet Duck

Ex2:The Matrix

-

- 3

- 5

- 7

-

- 0

- 0

- 7

-

- 9

- 1

- 1

Comments

- Comments/blank lines can go almost anywhere
- Must not be ambiguous with content
- Comments begin '#'

```
# comment before document
--- #DIRECTIVE # comment
foo: bar # inline comment
phone: number #555-1234
    ### Comment
fact: fiction
---
blue: bird
# Comment
```


Scalars

- A YAML scalar is strings, integers, dates and other atomic data types.
- YAML has many scalar forms.

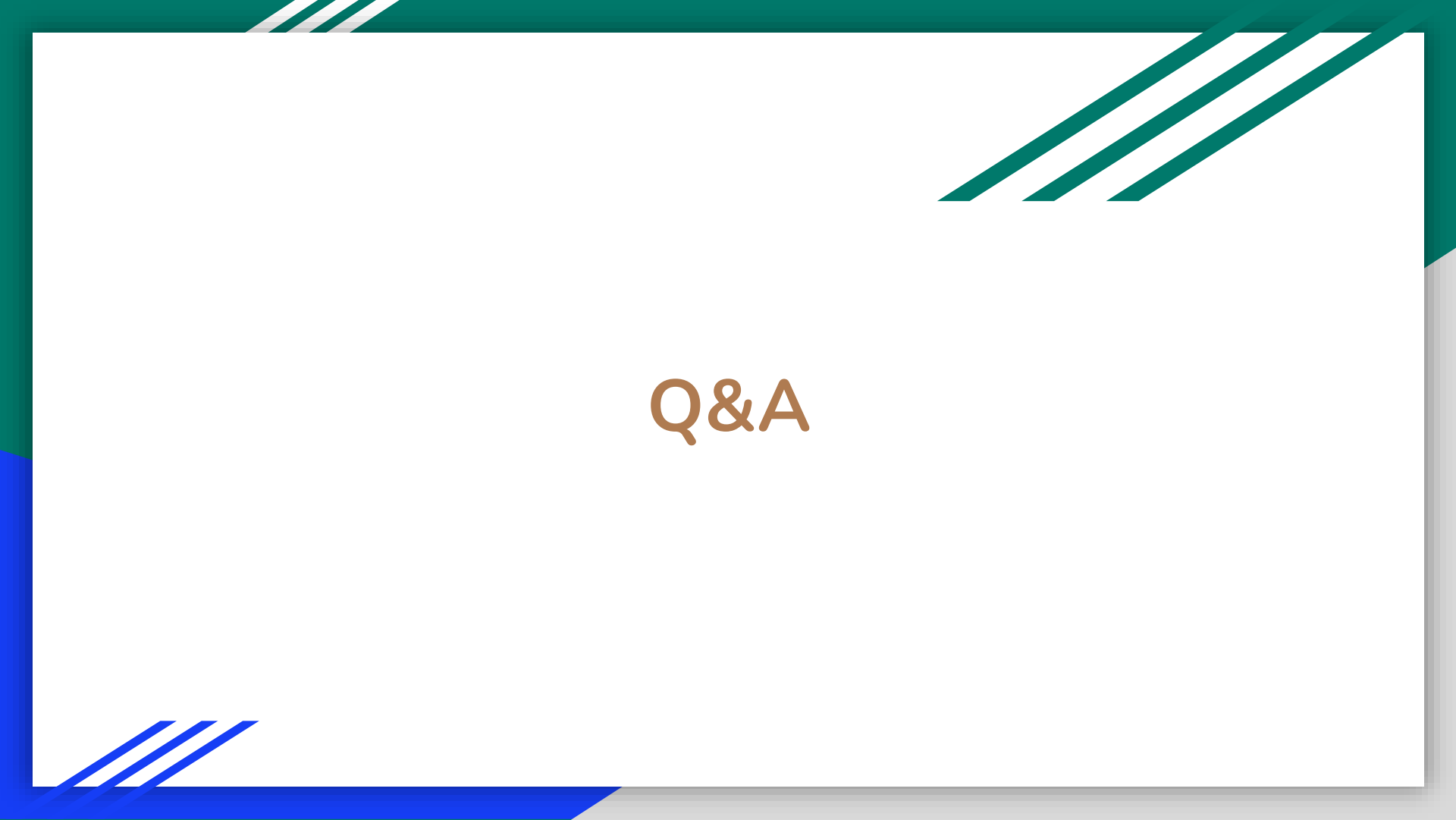
```
---
simple: look ma, no quotes
quoted:
  - 'Single quoted. Like Perl, no escapes'
  - "Double quotes.\nLike Perl, has escapes"
  - |
    A YAML block scalar.
    Much like Perl's
    here-document.
```

Scalars

Scalar content can be written in block form using a literal style (“|”) where all line breaks count. Or they can be written with the folded style (“>”) where each line break is folded to a space unless it ends an empty or a “more indented” line.

Example 2.16. Indentation determines scope

```
name: Mark McGwire
accomplishment: >
  Mark set a major league
  home run record in 1998.
stats: |
  65 Home Runs
  0.278 Batting Average
```



Q&A